David Gauthier

Paris, France · david.victor.gauthier@gmail.com
davidgaut.github.io · Google Scholar · github.com/davidgaut

Empirical macro-finance economist. I build novel datasets from unstructured sources — speeches, news archives, transcripts — to measure how information and technology propagate through the economy. Four refereed publications on credit, risk premia and the business cycle; research economist inside two central banks (Banque de France, Bank of England); retrained in data science at ENSAE after the Ph.D.

Positions

Research Economist, Banque de France — Financial Stability Directorate 2024 – present

Entity-level measurement of shock propagation and contagion in the banking system, used in policy analysis. Research on monetary policy, central bank communication and financial cycles.

Quantitative Research Intern, Capital Fund Management — Research Team 2022 (10 months)

End-to-end extraction of economic signal from central bank communication: corpus construction, NLP scoring, validation and deployment into decisions.

Research Economist, Bank of England — Modeling Team 2018 – 2021

Built and maintained the forecasting and modelling toolboxes (time-series, state-space, DSGE) used in the production forecast round. Research on business cycles, monetary economics and financial shocks; Brexit and unconventional monetary policy.

Ph.D. Student, Banque de France — Research Division & Paris School of Economics 2015 – 2018

Research — Working Papers

Authority, Attention, and the Origins of Monetary Policy Surprises 2026

Two novel datasets built for it: 4,177 Federal Reserve Board speeches and testimonies since 1961, scored into a policy-stance signal, and a press-attention panel matching every speech since 1992 to ∼100,000 Reuters articles (Factiva), alongside FOMC transcripts, voting records and real-time macroeconomic vintages. Shows that markets systematically discount policymaker communication: the discount predicts rate forecast errors, Treasury excess returns and the forward-guidance component of policy surprises — out of sample and under conservative inference, conditioning on the yield curve and a real-time macro panel.

Influence within the FOMC in progress

Identifies members' causal influence on policy decisions from the predetermined voting rotation of regional Fed presidents — a quasi-experimental design built from institutional rules.

Revisiting the Contribution of Fire-Sales to System-Wide Amplification in progress

A stress-testing model for the Eurozone built from entity-level balance-sheet data to measure the propagation of financial shocks across the system.

Research — Publications

Collateral Shocks 2022

with Y. Bécard — American Economic Journal: Macroeconomics

Collateral shocks — changes in the financial sector's ability to redeploy collateral — are the dominant source of U.S. business cycles. The estimated shock tracks measured lending standards and market sentiment.

Banks, Nonbanks, and Business Cycles 2023

with Y. Bécard — European Economic Review

A single risk-premium shock generates the comovement of euro-area macroeconomic and financial aggregates once banks pass risky loans to nonbanks, and drives business and financial cycles.

Slow Recoveries, Endogenous Growth and Macro-prudential Policy 2023

with D. Bonciani and D. Kanngiesser — Review of Economic Dynamics

Bank resilience raises the steady-state growth rate. Optimal capital ratio of 18%, with a welfare gain fourteen times larger than without endogenous growth. Open-source code.

Bank Shocks and the Debt Structure 2024

Macroeconomic Dynamics

Identifies bank credit-supply shocks from the composition of firms' debt — only bank shocks move loans and bonds in opposite directions. A third of output fluctuations; predicts the corporate bond spread.

Education

M.Sc. Data Science, ENSAE — Institut Polytechnique de Paris 2021 – 2022

Thesis: *Central Banks' Speeches: An NLP-Based Investment Strategy*.

Ph.D. Macroeconomics, Paris School of Economics & Panthéon-Sorbonne 2015 – 2019

Thesis: *Financial Stress and the Business Cycle*.

M.Sc. Economics (Research), Paris School of Economics & Panthéon-Sorbonne 2011 – 2013

Public Data & Code

Public datasets (Kaggle). Central bank speeches, eight institutions since 1997; FOMC press conferences, transcripts and audio since 2015; macroeconomic releases paired with consensus forecasts; UK Glassdoor reviews since 2008.

Code. github.com/davidgaut; Bank of England *Bank Underground* post.

Prediction Competitions

AMF — European taxonomy text classification, NLP	*top 1%*	2023
Natixis — asset returns from central bankers' speeches, NLP	*top 5%*	2022
Capital Fund Management — end-of-day stock auction volumes	*top 5%*	2021
Enefit — energy behaviour of prosumers	*top 5%*	2024
Optiver — realized volatility prediction	*top 10%*	2021

Skills

Programming. Python — scraping and parsing archives, ETL pipelines, feature construction, estimation, evaluation (pandas, numpy, scikit-learn, PyTorch, statsmodels); SQL, R, Matlab, Dynare, Git, LaTeX.
Methods. Causal inference (panel, event-study and quasi-experimental designs; structural identification); out-of-sample evaluation with HAC and bootstrap inference; real-time vintage data; Bayesian state-space and DSGE estimation; gradient boosting; transformer- and dictionary-based NLP for social science.
Data. Factiva / Dow Jones, Thomson Reuters Eikon, S&P Market Intelligence, FRED/ALFRED real-time vintages, federal funds futures, Blue Chip and SPF surveys.
Refereeing. Journal of Money, Credit and Banking · Quarterly Review of Economics and Finance · Macroeconomic Dynamics.
Teaching. Macroeconomics (undergraduate), Paris 1; Development Economics (graduate), CentraleSupélec.
Languages. French (native), English (fluent), Italian (beginner).